GOOGLE DEEPMIND AI HACKATHON • TRACK 2 (HARD TECH) & TRACK 3 (SOCIAL GOOD)

GeoRescue 3D

Real-Time Spatial Disaster Evacuation Simulator & AI Verification Engine

Gemini 4 Argon (1M Context, thinking_level="high")

Antigravity Sandbox Verification (antigravity-preview-09-2026)

Three.js 3D WebGL Digital Twin

FastAPI + NetworkX A* on Google Cloud Run

Live Google Cloud Run Production

https://georescue3d-464836279033.asia-northeast3.run.app/

Containerized FastAPI + Three.js 3D WebGL Digital Twin of the Seoul Gangnam Station Basin with real-time 1.04ms NetworkX A* verification.

Public GitHub Repository

https://github.com/kyungho-lee/20261016_Deepmind_AIHackathon

Includes 3-Team Competitive Tournament pipeline, Autonomous Sandbox Verifier, V-World/OSM Open GIS datasets, and HITL dispatch console.

Why 2D Flat Navigation & Unverified LLMs Fail in Flash Floods

1. Bowl-Shaped Gangnam Basin

Seoul's Gangnam Station intersection forms a steep topographical bowl (central underpass at **13.5m** surrounded by **38.5m–41.2m** hills).

During cloudbursts (`water_level = 2.0m`, flood plane `z = 18.5m`), stormwater rapidly submerges the central basin.

2. Deadly 2D Shortest Paths

Standard 2D map routers optimize strictly for horizontal distance (`850m`):
`GN_01` → `GN_02` → `GN_06` → `GN_08`

This leads citizens directly into the submerged **Gangnam Underpass (GN_02, z = 13.5m, water = 0.45m)**.

3. The GeoRescue 3D Solution

Combines **Gemini 4 Argon (1M Context)** with an isolated **Python NetworkX A* Sandbox Verifier**.

Guarantees every evacuation node satisfies `z_elevation > 18.5m` and `water_level_m ≤ 0.15m` before rendering on a 3D WebGL Digital Twin.

Gemini 4 Argon (1M) + Antigravity Autonomous Verification


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[1. GIS Data Pipeline (make_data.py)]  V-World 3D DEM (gangnam_3d_grid.csv) + OSM Graph (gangnam_graphify.json)
        |
        v
[2. Lead Orchestrator & Chief Jury]    gemini-4-argon (1M Context, thinking_level="high", background=True)
        |                              • Orchestrates 3-Team Tournament (Alpha / Beta / Gamma) & 100-pt Rubric
        v
[3. Autonomous Sandbox Verifier]       antigravity-preview-09-2026 (Isolated Linux Python NetworkX A* Sandbox)
        |                              • Iteration #1: Rejects GN_02 (13.5m ≤ 18.5m) -> Iteration #2: Verifies High
Ridge
        v
[4. Cloud Run Production WebApp]       FastAPI (app.py) + Three.js 3D WebGL + Google Maps + HITL Officer Approval
```

Agent Role	Model / Agent ID	Deliverable & Responsibility
1. Argon Chief Jury & Orchestrator	gemini-4-argon (high, 1M)	Multi-agent tournament orchestration, 3D physics code synthesis, 100-pt rubric evaluation
2. Autonomous Verification Agent	antigravity-preview-09-2026	Executes Python NetworkX A* verification & self-correction loop ([SUCCESS] Verified)
3. Fast Worker & Safety Guardrail	gemini-3.8-flash + 3.5-flash-lite	Sub-ms CSV/JSON validation, V-World/OSM ODbL license compliance & HITL officer gate

How Antigravity Detects & Bypasses Submerged Traps

Iteration #1 — Rejected by Sandbox Verifier

- **Candidate Route**: GN_01 (19.2m) → GN_02 (13.5m) → GN_06 (34.0m) → GN_08 (41.2m)
- **Horizontal Distance**: 850.0 m (Naive shortest path)
- **Sandbox Audit Result**: [FAIL / 4 CRITICAL VIOLATIONS]
- **Violation #1**: Node GN_02 elevation 13.50m ≤ 18.50m flood plane
- **Violation #2**: Surface water 0.45m > 0.15m safe pedestrian limit

Iteration #2 — Autonomous Self-Correction

- **Argon Self-Correction**: Blacklists GN_02, raises min_safe_z=16.0m, applies elevation bonus (2.0)
- **Verified Safe Route**: GN_01 (19.2m) → GN_04 (22.8m) → GN_05 (38.5m) → GN_06 (34.0m) → GN_08 Shelter (41.2m)
- **Min Elevation / Max Water**: 19.20m (> 18.5m) / 0.12m (≤ 0.15m)
- **Sandbox Output**: [SUCCESS] Verified by Python A* Algorithm

Terrain-Locked 3D WebGL Digital Twin & HITL Console

0.00m Elevation Sync

64 × 64 DEM

Unified 12-control-point IDW terrain interpolation across Python backend and Three.js frontend. Every road and green 3D evacuation tube hugs the exact slope with **2.5m contour lines**.

Real-Time Flood Slider

1.04 ms

Dragging the flood plane slider (`14.0m - 22.0m`) dynamically raises the 3D water mesh, updates the **Longitudinal Elevation Profile Chart**, and recalculates A* paths in `1.04ms`.

Google Maps + HITL Gate

100% Safe

Synchronized **WGS84 Coordinate Map**, one-click **Google Maps Walking Directions**, Open GIS attribution (V-World & OSM ODbL), and **Control Center Officer Approval**.

3-Team Competitive Tournament & Benchmark Results

Verification Dimension	Target Invariant / Threshold	Team Alpha (A*)	Team Beta (3D UI)	Team Gamma (Adopted Winner)
Min Route Elevation (`z_elevation`)	> 18.50 m	13.50 m (FAIL)	17.20 m (FAIL)	**19.20 m → 41.20 m (PASS)**
Max Surface Water Depth	≤ 0.15 m	0.45 m (FAIL)	0.28 m (FAIL)	**0.12 m (PASS)**
Golden Time Evacuation ETA	≤ 1,800 sec (30 min)	1,598.4 sec	1,412.0 sec	**1,524.6 sec (25.4 min — PASS)**
Argon Chief Jury Total Score	100-Point Rubric	47.0 / 100	44.0 / 100	**97.6 / 100 (ADOPTED)**

6. LIVE DEPLOYMENT & CONCLUSION

Ready for Real-World Urban Disaster Response

Try the Live 3D Simulation Now

Deployed on **Google Cloud Run** (asia-northeast3, Seoul):

https://georescue3d-464836279033.asia-northeast3.run.app/

Launch Live 3D WebGL Digital Twin ↗

Complete Open-Source Repository

All multi-agent pipelines, datasets, and verification logs:

https://github.com/kyungho-lee/20261016_Deepmind_AIHackathon

Open GitHub Repository ↗